PCA

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Suppose, we have

D1 (1000 columns)

D2 (200 columns)

Which gives highest accuracy?

Curse of Dimensionality.

* As no. of features increases there will be less model performance.
* Computational time is also high.
* Hard to visualize data.
* All the above things leads to overfitting.

How to overcome this?

1. Using Feature Selection → find correlation.
2. Using Feature extraction → extract new features from existing features.

How to do Feature Extraction? → using PCA.

PCA (Principal Component Analysis)

* It is used for dimensionality reduction & data compression.
* It is used for reducing number of features i.e dimension.

Eg:- If I have a dataset of 10 columns,

$C_1 \quad C_2 \quad . \quad . \quad . \quad . \quad . \quad . \quad C_8 \quad C_9 \quad C_{10}$

$\downarrow$ PCA

$D_1 \quad D_2 \quad D_3 \cdot D_4.$

* When we apply PCA on this data of 10 columns, PCA will convert higher dimension data (10 columns) into lower dimension data (4 columns)
* No data is lost in this process. Based on similarity of data

* No data is lost in this process. Based on similarity of data points, data is compressed into lower dimension.

* The new set of features created are called as Principal Components. ($D_1 D_2 D_3 D_4$)

Working of PCA

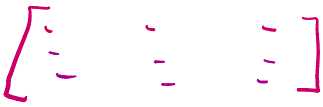
① Standardizing data

Converts all the features such that they have mean = 0, std = 1.

② Calculate covariance matrix

Covariance tells how features are related.

$X_1 \quad X_2 \quad X_3$

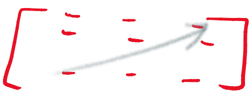


(3×3)

Covariance tells all these data points are related.

③ Calculate Eigen vectors & Eigen values.

Eigen vectors – tells us direction in which data is more similar.



Eigen values – will fetch all those data which are more similar & compress it into single Principal component.

How to select Optimal value for PCA?

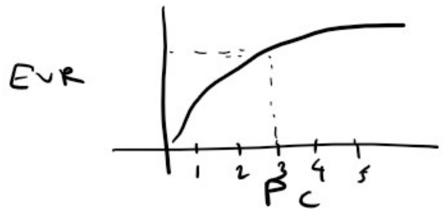
Explained Variance Ratio (EVR)

EVR tells us which PC's are showing how much variance for us whichever PC's are giving highest EVR has to be selected as optimal value for PCA.

Eg:- PC_1, PC_2

$$\boxed{\text{EVR of } PC_1 = \frac{\text{variance of } PC_1}{(\text{variance of } PC1) + (\text{variance of } PC2)}}$$

This needs to be plotted on Scree plot.



Select the PC, after which EVR value goes constant.